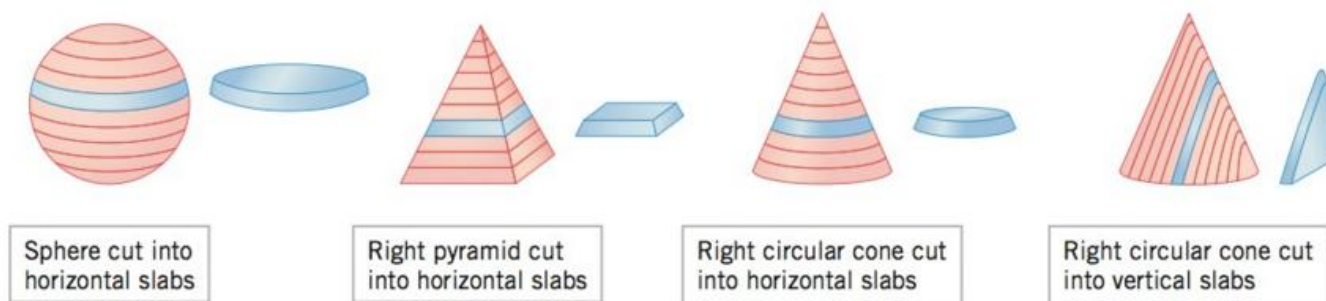


1.2 Volumes of Revolutions (Disk & Washer Method)

General Idea:

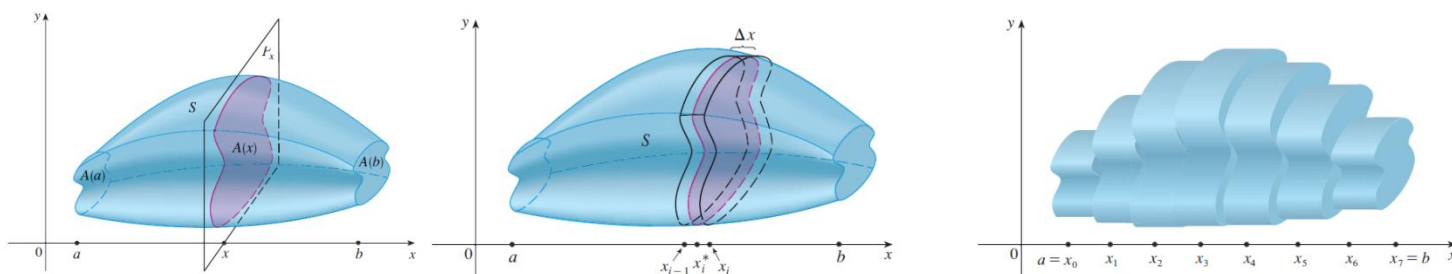
In this section we want to find a way to compute volumes of complicated solids using calculus. With this goal in mind we begin with a simple idea.

Notice we can estimate the volumes of three dimensional figures by adding up the volumes of “slices” or “slabs” of the figure. Some possible slices are pictured below.



Notice that as the number of slices n increases towards infinity the cross-sections become extremely thin and our approximation improves.

When we have more complicated solids, we will approximate the volume by using “cross sectional slabs” with uniform height and the same base shape on top and bottom. The next picture and definition make this idea precise.



Definition: Let S be a solid that lies between $x = a$ & $x = b$. If the cross-sectional area of S in the plane P_x , through x and perpendicular to the x -axis, is $A(x)$, where A is a continuous function, then the volume of S is

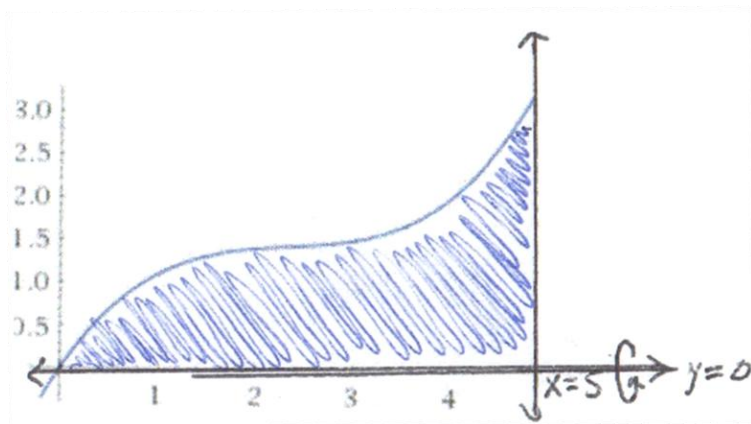
$$V = \lim_{n \rightarrow \infty} \sum_{i=1}^n A(x_i^*) \Delta x = \int_a^b A(x) dx$$

Note: If the solid is bounded by two numbers on the y axis, the definition becomes $V = \int_a^b A(y) dy$.

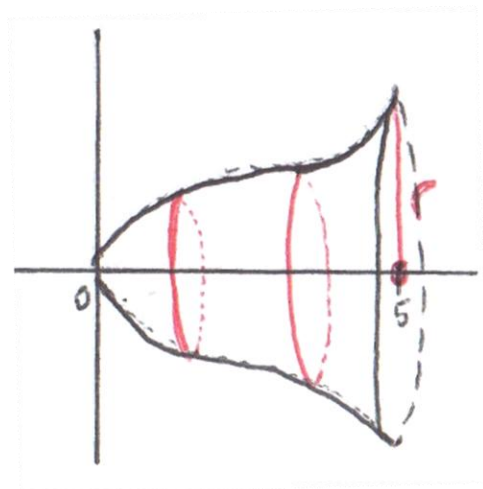
Our first application of this definition will be on solids of revolution. The idea behind solids of revolution is that we will rotate a region about the x or y axis and want to know the volume of the solid obtained.

Example 1: Consider the function $f(x) = 0.1x^3 - 0.7x^2 + 1.7x$

- a) Suppose we consider the region bounded by $f(x)$ and the lines $y=0$ & $x=5$.



- b) Imagine that we revolve the area pictured to the left around the x -axis. Sketch a picture of the solid generated from performing this revolution.



- c) Imagine if we make vertical cuts into our 3-dimensional figure. What is the shape of the cross-sections?

If we look at the red sections above we may suspect that these cross-sections are circles.

Indeed these sections are circles because the height of the curve at any point remains constant as the revolution is completed.

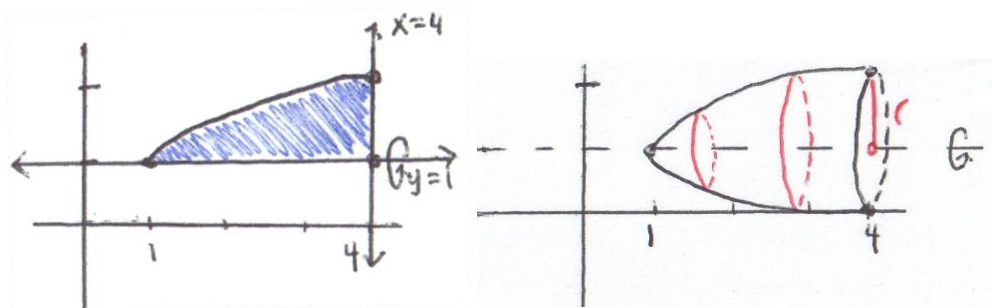
Now we can use our definition of volume to find the volume for our 3-dimensional solid.

- d) Set up an integral to find the volume of the 3-dimensional solid pictured in part (b).

$$V = \int_0^5 \pi [f(x)]^2 dx = \int_0^5 \pi [0.1x^3 - 0.7x^2 + 1.7x]^2 dx \approx 44.973 \text{ (units cubed)}$$

Note: From a Riemann sum point of view, we are really approximating the volume by using disks (i.e. right circular cylinders with very small height.)

Example 2: Find the volume of the solid obtained by rotating the region between the curves $y = \sqrt{x}$ and $y = 1$ from $x = 1$ to $x = 4$ about the line $y = 1$.



We can see that the cross sections of the solid are circles with a radius of $\sqrt{x} - 1$ (since we are revolving about the line $y = 1$). So we write:

Setup:

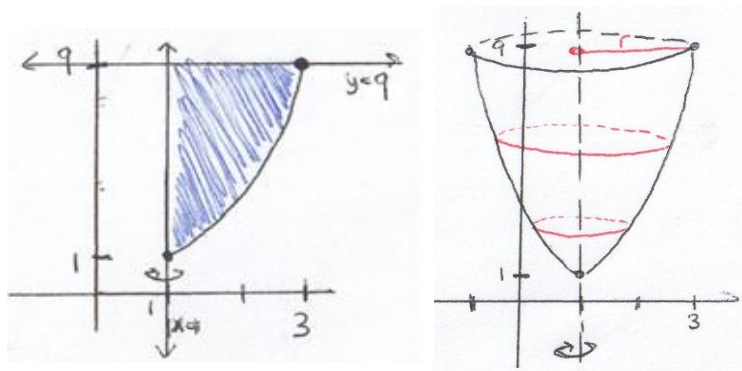
$$V = \int_1^4 \pi [\sqrt{x} - 1]^2 dx$$

Evaluate:

$$V = \int_1^4 \pi [\sqrt{x} - 1]^2 dx = \pi \int_1^4 (x - 2\sqrt{x} + 1) dx = \pi \left(\frac{1}{2}x^2 - \frac{4}{3}x^{3/2} + x \right) \Big|_1^4 = \pi \left[\frac{16}{2} - \frac{32}{3} + 4 \right] - \pi \left[\frac{1}{2} - \frac{4}{3} + 1 \right] = \frac{7\pi}{6}$$

Example 3: Find the volume of the solid obtained by rotating the region bounded by $y = x^2$ & $y = 9$ (which lies to the right of $x = 1$) about the line $x = 1$.

We notice that the cross sections of the solids are again circles, but they are oriented differently than the previous two problems. Since we rotated around vertical axis our circles are stacking vertically. Therefore, we need to set up an integral with respect to y .



So, we solve for x : $y = x^2 \Rightarrow x = \sqrt{y}$.

Our circles therefore have a radius of $(\sqrt{y} - 1)$. So we write

Setup:

$$V = \int_1^9 \pi [\sqrt{y} - 1]^2 dy$$

Evaluate:

$$V = \int_1^9 \pi [\sqrt{y} - 1]^2 dy = \pi \int_1^9 (y - 2\sqrt{y} + 1) dy = \pi \left(\frac{y^2}{2} - \frac{4}{3}y^{3/2} + y \right) \Big|_1^9 = \pi \left(\frac{81}{2} - 36 + 9 \right) - \pi \left(\frac{1}{2} - \frac{4}{3} + 1 \right) = \frac{40\pi}{3}$$

Note: In all three examples we have seen, from a Riemann sum point of view we are approximating the volumes with disks. The general principle at play is stated below.

Disk Method

Volume of a Solid of Revolution (w/ a horizontal axis of rotation)

If S is a solid that lies between $x=a$ & $x=b$ with circular cross-sections which are perpendicular to the

x -axis, then the volume of S is given by $V = \int_a^b \pi r^2 dx$

where r is a function of x which represents the radius of the circular cross-section at x .

Volume of a Solid of Revolution (w/ a vertical axis of rotation)

If S is a solid that lies between $y=a$ & $y=b$ with circular cross-sections which are perpendicular to the

y -axis, then the volume of S is given by $V = \int_a^b \pi r^2 dy$

where r is a function of y which represents the radius of the circular cross-section at y .

Example 4: Find the volume of the solid obtained by rotating the region bounded by $y=3x$, $y=12-3x$, & $y=0$ about the x -axis.

Since the disks stack up left to right, this problem should be done in terms of x .

However we see that the radius of the disks is determined by two different functions (switching over when $x=2$ so we will need 2 separate integrals.

Our cross sections are circles with radius $3x$ from $[0, 2]$ and $12-3x$ from

$[2, 4]$.

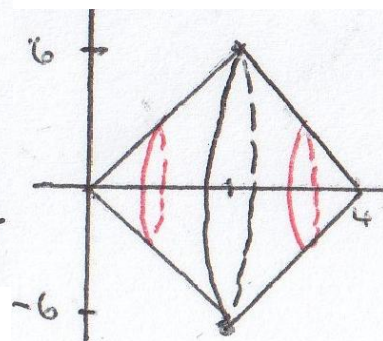
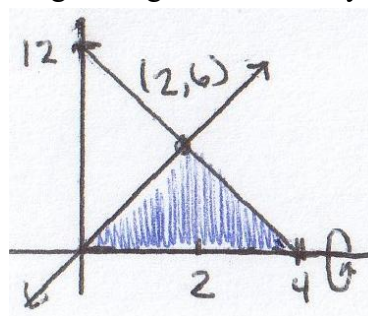
So we write:

Setup:

$$V = \int_0^2 \pi [3x]^2 dx + \int_2^4 \pi [12-3x]^2 dx$$

Evaluate:

$$\begin{aligned} V &= \int_0^2 \pi [3x]^2 dx + \int_2^4 \pi [12-3x]^2 dx = \pi \int_0^2 9x^2 dx + \pi \int_2^4 [9x^2 - 72x + 144] dx \\ &= \pi (3x^3) \Big|_0^2 + \pi (3x^3 - 36x^2 + 144x) \Big|_2^4 \\ &= \pi [24 - 0] + \pi [192 - 168] = 48\pi \end{aligned}$$

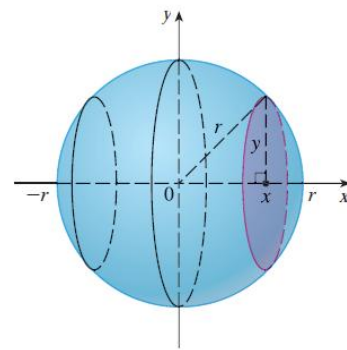


Example 5: Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.

Since the disks stack up left to right, this problem should be done in terms of x .

We notice that the radius of each disk is determined by the circle $x^2 + y^2 = r^2$ which implies that $y = \pm\sqrt{r^2 - x^2}$.

So we then see that our cross sections are circles with radius $y = \sqrt{r^2 - x^2}$. Therefore we write:



Setup:

$$V = \int_{-r}^r \pi \left[\sqrt{r^2 - x^2} \right]^2 dx = 2\pi \int_0^r (r^2 - x^2) dx$$

Evaluate:

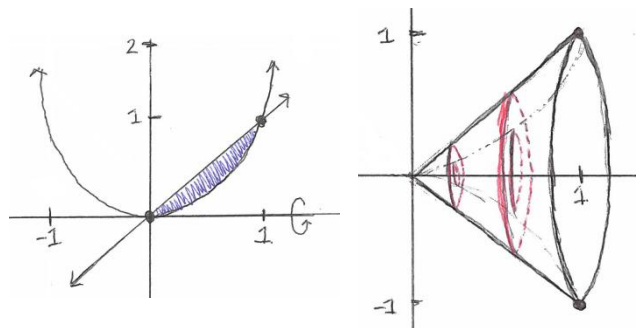
$$\begin{aligned} V &= 2\pi \int_0^r (r^2 - x^2) dx = 2\pi \left(r^2 x - \frac{1}{3} x^3 \right) \Big|_0^r \\ &= 2\pi \left[r^2 r - \frac{1}{3} r^3 \right] - 2\pi [0] \\ &= 2\pi \left[r^3 - \frac{1}{3} r^3 \right] \\ &= 2\pi \left[\frac{2}{3} r^3 \right] \\ &= \frac{4\pi}{3} r^3 \end{aligned}$$

How cool is that!?

Example 6: Find the volume when the region enclosed by the curves $y = x$ & $y = x^2$ is rotated about the x -axis.

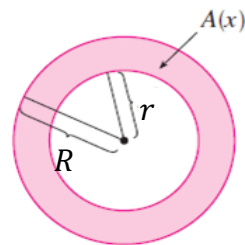
Since the cross-sections stack up left to right, this problem should be done in terms of x .

But how can we do this? The cross-sections are not a very familiar shape like a circle or a square. What is the area of this cross-section?



Our cross-sections look like washers or one circle being “punched out” of another. To find the area of this section we can compute:

$$A = \pi R^2 - \pi r^2 \quad \text{or} \quad A = \pi(R^2 - r^2)$$



Since the washers stack up left to right, this problem should be done in terms of x .

We can see that our washers have a large radius given by x and a small radius of x^2 . Therefore we write:

Setup:

$$V = \int_0^1 \left[\pi(x)^2 - \pi(x^2)^2 \right] dx = \int_0^1 \pi \left[(x)^2 - (x^2)^2 \right] dx$$

Evaluate:

$$V = \int_0^1 \pi \left[(x)^2 - (x^2)^2 \right] dx = \pi \int_0^1 [x^2 - x^4] dx = \pi \left(\frac{1}{3}x^3 - \frac{1}{5}x^5 \right) \Big|_0^1 = \pi \left[\frac{1}{3} - \frac{1}{5} \right] - \pi[0] = \frac{2\pi}{15}$$

The general principle at play in this example is stated below.

Washer Method

Volume of a Solid of Revolution (w/ a horizontal axis of rotation)

If S is a solid that lies between $x=a$ & $x=b$ with washer cross-sections which are perpendicular to the

x -axis, then the volume of S is given by $V = \int_a^b (\pi R^2 - \pi r^2) dx$

where R is the outer radius, r is the inner radius and both R and r are functions of x .

Volume of a Solid of Revolution (w/ a vertical axis of rotation)

If S is a solid that lies between $y=a$ & $y=b$ with washer cross-sections which are perpendicular to the

y -axis, then the volume of S is given by $V = \int_a^b (\pi R^2 - \pi r^2) dy$

where R is the outer radius, r is the inner radius and both R and r are functions of y .

Example 7: Find the volume when the region enclosed by the curves $y = x$ & $y = x^2$ is rotated about the line $y = 2$.

Since the washers stack up left to right, this problem should be done in terms of x .

We can see that $R = 2 - x^2$ and $r = 2 - x$. Therefore we have:

Setup:

$$V = \int_0^1 \left[\pi(2 - x^2)^2 - \pi(2 - x)^2 \right] dx = \pi \int_0^1 \left[(2 - x^2)^2 - (2 - x)^2 \right] dx$$

Evaluate:

$$\begin{aligned} V &= \pi \int_0^1 \left[(2 - x^2)^2 - (2 - x)^2 \right] dx = \pi \int_0^1 \left[4 - 4x^2 + x^4 - (4 - 4x + x^2) \right] dx = \pi \int_0^1 \left[x^4 - 5x^2 + 4x \right] dx \\ &= \pi \left(\frac{1}{5}x^5 - \frac{5}{3}x^3 + 2x^2 \right) \Big|_0^1 = \pi \left[\frac{1}{5} - \frac{5}{3} + 2 \right] - \pi[0] = \frac{8\pi}{15} \end{aligned}$$

Note: This example shows us how important it is to always draw/measure our radius back to our original functions.

Example 8: Find the volume when the region enclosed by the curves $y = x$ & $y = x^2$ is rotated about the line $x = -1$.

Since the washers stack up top to bottom, this problem should be done in terms of y .

We must solve our equations for x :

$$y = x \text{ \& } y = x^2 \Rightarrow x = y \text{ \& } x = \sqrt{y}$$

We can see that $R = \sqrt{y} - (-1) = \sqrt{y} + 1$ and

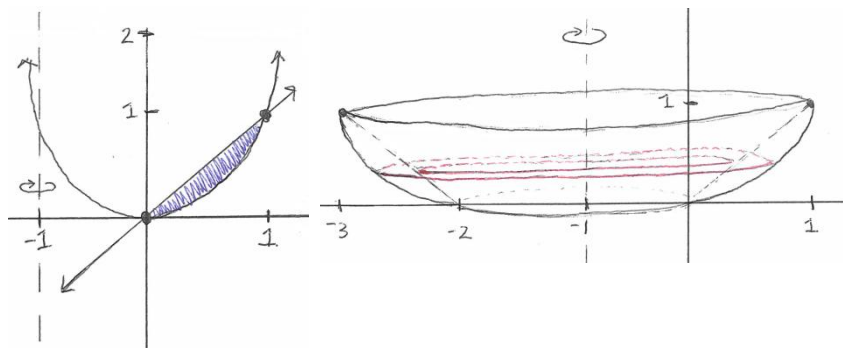
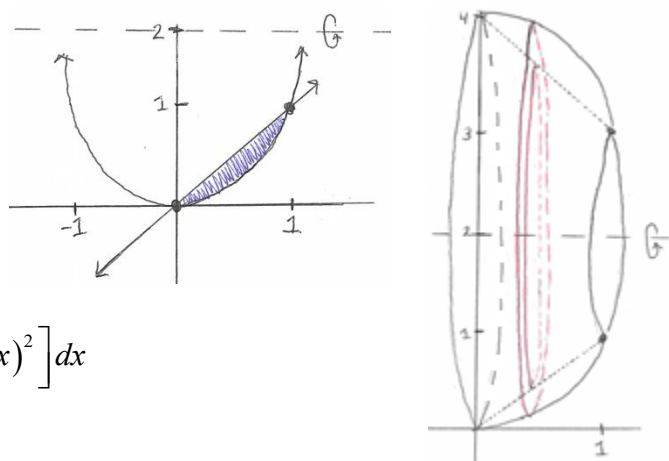
$r = x - (-1) = x + 1$. Therefore we have:

Setup:

$$V = \int_0^1 \left[\pi(\sqrt{y} + 1)^2 - \pi(y + 1)^2 \right] dy = \pi \int_0^1 \left[(\sqrt{y} + 1)^2 - (y + 1)^2 \right] dy$$

Evaluate:

$$\begin{aligned} V &= \pi \int_0^1 \left[(\sqrt{y} + 1)^2 - (y + 1)^2 \right] dy = \pi \int_0^1 \left[(y + 2\sqrt{y} + 1) - (y^2 + 2y + 1) \right] dy = \pi \int_0^1 \left[-y^2 - y + 2\sqrt{y} \right] dy \\ &= \pi \left(-\frac{1}{3}y^3 - \frac{1}{2}y^2 + \frac{4}{3}y^{3/2} \right) \Big|_0^1 = \pi \left[-\frac{1}{3} - \frac{1}{2} + \frac{4}{3} \right] - \pi[0] = \frac{\pi}{2} \end{aligned}$$



Solids Built From Bases

Example 9: Let the region formed by the function $y = x^2$ and the line $y = 4$ form the base of a solid for which the cross-sections parallel to the x -axis are squares. Find the volume of the resulting figure.

Find a formula for the area of the cross-section:

Half the length of the square is determined by the x -coordinate of the function $y = x^2$. We can solve for the x -coordinate (we take the positive x -value) $x = \sqrt{y}$

Therefore the total side length is $s = 2\sqrt{y}$.

The area of the cross-section is $A = s^2 = 4y$.

Therefore we write that:

Setup:

$$V = \int_0^4 4y \, dy$$

Evaluate:

$$V = \int_0^4 4y \, dy = 2y^2 \Big|_0^4 = 32 - 0 = 32$$

